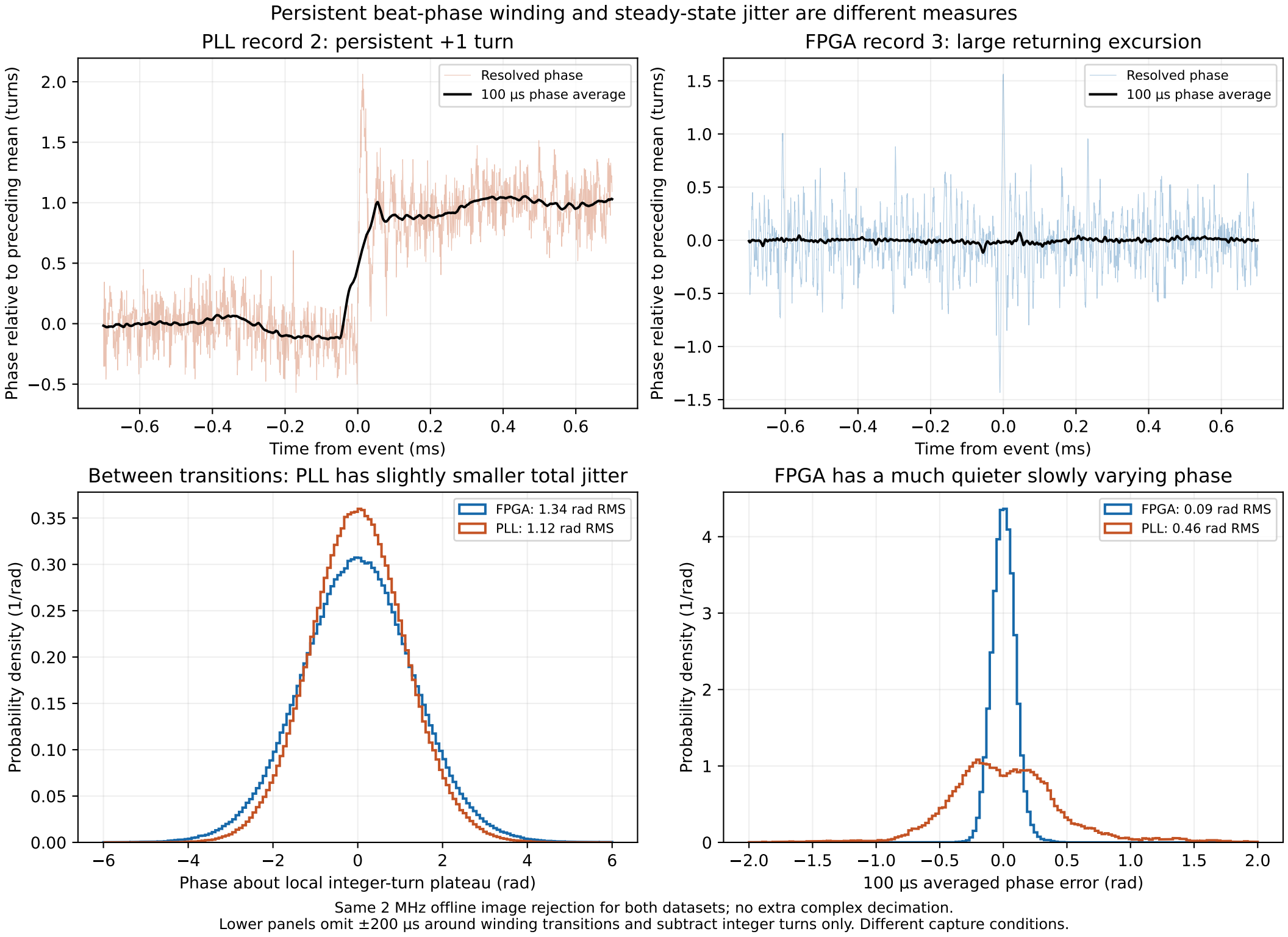

Persistent beat-phase winding and steady-state jitter are different measures
PLL record 2: persistent +1 turn
FPGA record 3: large returning excursion
Resolved phase
100 µs phase average
Phase relative to preceding mean (turns)
Time from event (ms)
Between transitions: PLL has slightly smaller total jitter
FPGA has a much quieter slowly varying phase
FPGA: 1.34 rad RMS
PLL: 1.12 rad RMS
FPGA: 0.09 rad RMS
PLL: 0.46 rad RMS
Probability density (1/rad)
Phase about local integer-turn plateau (rad)
100 µs averaged phase error (rad)
Same 2 MHz offline image rejection for both datasets; no extra complex decimation.
Lower panels omit ±200 µs around winding transitions and subtract integer turns only. Different capture conditions.